

Dataset Analysis Approach

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 **sicor_municipal_co... .csv**
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To begin analyzing the dataset `sicor_municipal_contracts.csv`, I will first load the data and examine its structure, including column names, data types, and key statistical properties. This initial exploration will help identify the dataset's characteristics and inform how to address specific analytical questions effectively.

Step 1: Load and Inspect the Dataset

I will use Python's `pandas` library to load the dataset and inspect its contents.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows:")
13 print(data.head())
14
15 # Display summary statistics for numerical columns
16 print("\nSummary Statistics:")
17 print(data.describe())
```

Step 2: Identify Key Characteristics

From the output of the above code, I will:

1. List the column names and their data types.
2. Check for missing values in each column.
3. Identify numerical and categorical variables.
4. Summarize any notable patterns or trends in the data.

Once the dataset's structure is understood, I can proceed to address a specific analytical question using the data. Please provide the analytical question you would like me to answer, and I will craft a solution accordingly.

How can the distribution of investment values among funding, investment, commercialization, and industrialization be optimized to maximize the invested value? Define three options to address this issue and select the best one.

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To address the question of optimizing the distribution of investment values among funding, investment, commercialization, and industrialization to maximize the invested value, we need to analyze the dataset and propose actionable strategies. Here's how we can approach this:



Step 1: Understand the Dataset

From the dataset `sicor_municipal_contracts.csv`, we will:

1. Identify columns related to funding, investment, commercialization, and industrialization.
2. Analyze the distribution of values across these categories.
3. Determine any correlations or dependencies between these variables.

Assuming the dataset contains relevant columns (e.g., `funding`, `investment`, `commercialization`, `industrialization`), we will proceed with the analysis.

Step 2: Define Optimization Options

Based on the data, we will propose three optimization strategies:

- Option 1: Reallocate Resources Based on ROI**
Shift funds toward categories that historically yield the highest return on investment (ROI). This requires identifying which category has the highest ROI based on past performance.
- Option 2: Equalize Investment Across Categories**
Distribute resources equally among all categories to ensure balanced growth and mitigate risks associated with over-investing in a single area.
- Option 3: Prioritize High-Impact Categories**
Focus on categories with the largest absolute impact on total invested value, even if their ROI is moderate. This strategy prioritizes scale over efficiency.

Step 3: Evaluate and Select the Best Option

Using the dataset, we will calculate metrics such as total invested value, ROI, and impact for each category. Then, we will simulate the outcomes of each option and select the one that maximizes the invested value.

Python Code Implementation

Below is the Python code to analyze the dataset and evaluate the three options:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Inspect relevant columns
8 relevant_columns = ['funding', 'investment', 'commercialization', 'industrialization']
9 if not all(col in data.columns for col in relevant_columns):
10     raise ValueError("Dataset does not contain the required columns.")
11
12 # Calculate total invested value and ROI for each category
13 data['total_invested'] = data[relevant_columns].sum(axis=1)
14 data['roi_funding'] = data['funding'] / data['total_invested']
15 data['roi_investment'] = data['investment'] / data['total_invested']
16 data['roi_commercialization'] = data['commercialization'] / data['total_invested']
17 data['roi_industrialization'] = data['industrialization'] / data['total_invested']
18
19 # Summarize ROI and total invested value
20 roi_summary = data[['roi_funding', 'roi_investment', 'roi_commercialization', 'roi_industrialization']]
21 total_invested_summary = data[relevant_columns].sum()
22
23 print("ROI Summary:")
24 print(roi_summary)
25 print("\nTotal Invested Value Summary:")
26 print(total_invested_summary)
27
28 # Option 1: Reallocate Resources Based on ROI
29 # Allocate more resources to the category with the highest ROI
30 best_roi_category = roi_summary.idxmax()
31 option_1_allocation = {col: 0 for col in relevant_columns}
32 option_1_allocation[best_roi_category] = 1 # Allocate 100% to the best ROI category
33
34 # Option 2: Equalize Investment Across Categories
35 option_2_allocation = {col: 0.25 for col in relevant_columns}
36
37 # Option 3: Prioritize High-Impact Categories
38 # Allocate more resources to the category with the largest total invested value
39 best_impact_category = total_invested_summary.idxmax()
40 option_3_allocation = {col: 0 for col in relevant_columns}
```

```

41 option_3_allocation[best_impact_category] = 1 # Allocate 100% to the best impact category
42
43 # Simulate outcomes for each option
44 v def simulate_outcome(allocation, roi_summary):
45     return sum(allocation[col] * roi_summary[col.split('_')[1]] for col in allocation)
46
47 option_1_outcome = simulate_outcome(option_1_allocation, roi_summary)
48 option_2_outcome = simulate_outcome(option_2_allocation, roi_summary)
49 option_3_outcome = simulate_outcome(option_3_allocation, roi_summary)
50
51 # Compare outcomes and select the best option
52 v outcomes = {
53     "Option 1 (ROI-Based)": option_1_outcome,
54     "Option 2 (Equal Allocation)": option_2_outcome,
55     "Option 3 (Impact-Based)": option_3_outcome
56 }
57 best_option = max(outcomes, key=outcomes.get)
58
59 print("\nSimulated Outcomes:")
60 v for option, outcome in outcomes.items():
61     print(f"{option}: {outcome:.2f}")
62
63 print(f"\nBest Option: {best_option}")

```

Explanation of the Code

- Data Preparation :** The code identifies relevant columns (`funding` , `investment` , `commercialization` , `industrialization`) and calculates ROI and total invested value for each category.
- Option Evaluation :** Three allocation strategies are defined:
 - Option 1 :** Allocates 100% of resources to the category with the highest ROI.
 - Option 2 :** Allocates resources equally across all categories.
 - Option 3 :** Allocates 100% of resources to the category with the largest total invested value.
- Outcome Simulation :** The code simulates the outcomes of each option by calculating the weighted ROI based on the allocation.
- Best Option Selection :** The option with the highest simulated outcome is selected as the best strategy.

Expected Output

The output will include:

- A summary of ROI and total invested value for each category.
- Simulated outcomes for each option.
- The best option based on the simulation.

Conclusion

By analyzing the dataset and simulating outcomes, we can determine the optimal strategy for distributing investment values.

